

JOSEPH BINGHAM

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Research Statement

Topological representation learning and unsupervised methods, especially SOMs, spanning the theory of what learned representations encode and its application to fairness auditing, scientific discovery, and efficient learning. Bridging methodological work with real-world data in biology and clinical decision-making, including wet-lab experimental validation.

Research Interests

- Topological and geometric representation learning, especially with Self-Organizing Maps
- Auditing learned representations for fairness and sensitive-attribute leakage
- Structure-guided learning: label-free adaptation, compression, and cost-aware acquisition
- Graph-based and discrete mathematical foundations of learning

Publications

Journal Articles:

Fine-Pruning: A Biologically Inspired Algorithm for Personalization of Machine Learning Models [Joseph Bingham](#), [Saman Zonouz](#), [Dvir Aran](#) *Cell: Patterns*, Mar. 2025

Peer-Reviewed Conference Papers:

SOMnibus: Recovering Underlying Sensitive Attributes with Self-Organizing Maps

[Joseph Bingham](#), [Netanel Arussy](#), [Dvir Aran](#) *ICLR 2026 Workshop AFAA* Selected Poster

A Framework for Aligning Human Linguistics and AI Perception [Joseph Bingham](#) *ICLR 2026 Workshop HCAIR*

Genes Are Not Words: Dependency-Aware Masking for Single-Cell Foundation Models

[Alon Hacohen](#), [Joseph Bingham](#), [Dvir Aran](#) *ICLR 2026 Workshop Gen²* Top 10%

Guide-Guard: Off-Target Predicting in CRISPR Applications in the Genomic Domain

[Joseph Bingham](#), [Netanel Arussy](#), [Saman Zonouz](#) *IDEAL 2022*

LegoNet: Memory Footprint Reduction Through Block Weight Clustering [Joseph Bingham](#), [Noah Green](#), [Saman Zonouz](#) *IEEE DASC 2022 (20th year)* Oral (Top 10%)

Bonsai: A Framework for Convolutional Neural Network Acceleration Using Criterion-Based Pruning [Joseph Bingham](#), [S. Helmich](#) *MDML 2021*

Under Review:

SOMtime the World Ain't Fair: Violating Fairness Using Self-Organizing Maps

[Joseph Bingham](#), [Netanel Arussy](#), [Dvir Aran](#) *AAAI 2027*

LEGOLM: Structured Weight Sharing for LLMs [Joseph Bingham](#) *AAAI 2027*

Activation-Energy Pruning for Spiking Neural Networks: Unsupervised Personalization via Spike-Count Saliency [Joseph Bingham](#) *Neural Networks*

TRACE: Transparent Retrieval for Abstract Concept Evaluation [Joseph Bingham](#) *EACL 2027*

A compartment-adjustment framework reclassifies *FOXC1* as a stromal-vascular readout in Luminal A breast cancer and recasts the basal-lineage “Centaur” signal as a tumour-population axis [Daniel Yehoshua](#), [Joseph Bingham](#) *British Journal of Cancer*

EIG-Cost: Cost-Aware Active Feature Acquisition for Clinical Differential Diagnosis on MIMIC IV Data [Joseph Bingham](#), [Netanel Arussy](#) *JAMIA*

A Multimodal Framework for Aligning Human Linguistic Descriptions with Visual Perceptual Data [Joseph Bingham](#) *Cognitive Computation*

CorrMask: Enabling Single-Cell Foundation Models for Scarce Data via

Dependency-Aware Masking [Alon Hacohen](#), [Joseph Bingham](#), [Dvir Aran](#) *MLCB 2027*

Education

Israel Institute of Technology

Post Doctoral Fellow Biology

Haifa, IL

Aug. 2023 – present

- **Dvir Aran Lab**
- **Research Focus:** Using explainable AI techniques to explore biological/genetic data.
- Developing an independent research agenda on biologically grounded representation learning and efficiency working in and with both dry and wet labs.
- Utilizing Self Organizing Maps (SOM's) to create topological embeddings of latent genetic features present in cancer cells. Working with datasets ranging from 3,000 to 30,000,000 cells.
- Leading masters and doctoral students in their explorations and research as co-advisor, especially in computational aspects of their research/theses.
- Running wet lab experiments using state-of-the-art DNA/RNA sequencing techniques.

Rutgers University

Ph.D. Electrical and Computer Engineering

New Brunswick, NJ

Jan. 2021 – June 2023

- **Thesis: Accessible Artificial Intelligence**
- Advisors: Prof. Saman Zonouz, Dr. Eric Davis, Prof. Bo Yuan, Prof. Athina Petropulu
- Researching neural network acceleration techniques.
- Creating methods for symbiotic intelligence leveraging ontologies.

Rutgers University

M.S. Electrical and Computer Engineering

New Brunswick, NJ

Jan. 2019 – Jan. 2021

- **Thesis: Fine-Pruning: Fine Tuning Using Model Compression**
- Created and lead *Capture the Drone*, a competition for undergraduates to compete in hacking.

Iowa State University

B.S. Mathematics (cum laude)

Ames, IA

Aug. 2016 – May 2018

- **Thesis:** High Performance Genetic Algorithms for Steganalysis

Grants, Fellowships, Honors, and Awards

High Impact Researcher Postdoctoral Fellowship

Technion, 2023-present

Postdoctoral Fellowship

Misrad haAliyah, 2023-present

Up and Coming Researcher

Center for Integration in Science, 2023

NSF GAANN Fellowship

Rutgers University, 2019-2021

1st Place Prize

ISEAGE CDC, 2019

Thomas J. Miller Fellowship

Iowa State University, 2018

Grant for promising research assistants

Iowa State University, 2018

Eta Sigma Phi honors

Iowa State University, 2017

Presidential Scholarship

Iowa State University, 2016-2018

Cardinal Leadership Scholarship

Iowa State University, 2016-2018

Dean's List Honors

Iowa State University, 2016-2018

Skills

Languages: Python, C/C++, Java, SQL, Bash, Julia, R, NASM

Human Languages: English (native), French (C3), Latin (fluent), Spanish (B2), German (B2), Hebrew (conversational)

Developer Tools: Keras, TensorFlow, Pytorch, Jupyter Notebooks, SciPy, ScanPy, Seurat, Databrick, Git, DataBricks, Google Cloud Platform, Amazon AWS, VMware, Virtual Box

Wet Lab: Ficoll fixation of PBMC samples, 10x gem x protocol (from scrolls and fresh), Scale bioscience single cell library protocol, 10x single cell library protocol, Trained to handle/clean/store biological and chemical hazards

Relevant Experience

Researcher

Sep 2018 – Sep 2023

John Deere

Des Moines, IA/Champaign, IL (Hybrid)

- Created *Bonsai*, a keras based structural pruning framework, which reduces the complexity of real models by over 70% and increases the accuracy by 2% using novel pruning criterion.
- Worked with AWS instances for large scale data migration for large scale NN pipelines.
- Utilized parallelization to train Keras models to solve complex agricultural modeling problems.

Research Intern

May 2022 – Oct. 2022

Galois

Arlington, VA

- Worked with non-academic research scientists to create solutions to abstract problems.
- Collaborated with remote team members to solve problems in human perception and communication using web scraping and data mining.

Researcher

Jan. 2019 – Jan. 2021

New York Genome Center

New York City, NY

- Created *Guide-Guard*, a CNN solution in Keras that classifies gRNA sequences as safe for CRISPR CAS13 based applications with an accuracy of over 85%, 10% better than previous methods
- Collaborated with domain scientist to use statistical methods to solve biological based problems.

Teaching

Data Science for Biological & Medical Applications

Semester B 2024 - Present

Lecturer

Technion University

- Designed a course from scratch that addresses a diverse background of students from pure biology to computer science about intersectional topics while maintaining engagement.
- Taught students with no background in programming how to implement complex models in python using a variety of packages and environments.
- Acted as full lecturer, coordinating student, TA's, and administration to achieve learning goals.

Advanced Topics: Reverse Engineering & Malware Analysis

Spring 2019

Assistant Lecturer

Rutgers University

- Taught students how to apply the fundamental principles of dissecting malware, vulnerability finding/defense, and cyber attack triage, become aware of limitations of existing defense mechanisms and how to avoid them Study cutting-edge research publications on these topics
- Had students engage in critical discussion around key research topics in software security and forensics by having them propose solutions to open-ended research problems, and implement novel prototype solutions.

COM S 230/330 Discrete Computational Structures

Fall 2018

Teaching Assistant

Iowa State University

- Concepts in discrete mathematics as applied to computer science. Logic, set theory, functions, relations, combinatorics, discrete probability, graph theory, and number theory.
- Proof techniques, induction, and recursion.

Professional Service

Reviewer

2019 - Present

ICLR '26, KDD '26, ICASSP '25-26, HIP Fellows, '24 ICML '23, AAAI '23, IEEE DASC '23

Various

Approximating Human-Like Synaptic Pruning for Personalizing NNs

Galois Inc.

Invited Talk

2023

Cyber Security: Hacking from Micro-Processors to Buildings

IEEE S & P, Rutgers

Keynote Presenter

2020

Vulnerabilities of Modern Drone Systems

CyberSec

Guest Lecturer

2019

Industrial Applications of Statistical Principles

Mathematics and Statistics Society, ISU

Invited Speaker

2018